2022B F=ma Exam: Problem 9

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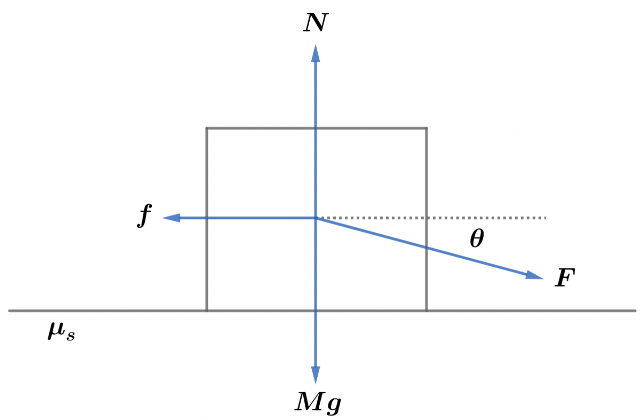

Balancing forces in the horizontal direction,

$$F\cos\theta = f = \mu_s N$$

Balancing forces in the vertical direction,

$$N = Mg + F\sin\theta$$

Substituting,

$$F\cos\theta = \mu_s(Mg + F\sin\theta)$$
$$F(\cos\theta - \mu_s\sin\theta) = \mu_s Mg$$
$$F = \frac{\mu_s Mg}{\cos\theta - \mu_s\sin\theta} \approx 203\,\mathrm{N}$$

so the answer is $\boxed{\text{C}}$.